

Grade 6 Accelerated
Unit 12
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

1. How much money would an investment of \$3,875 in a 3-year period and 4% simple interest rate makes?
 $3875 \cdot 0.04 \cdot 3 = 465$
The investment would make \$465 in interest.

2. \$560 are put into a savings account. The account has a 6% simple interest rate. How much money would the account have by the end of 3-years?
 $560 \cdot 0.06 \cdot 3 = 100.8$ $100.8 + 560 = 660.8$
The account would have \$660.80 in it.

3. How much money would an investment of \$7,730 during a 3-year period with a 4% simple interest rate make?
 $7730 \cdot 3 \cdot 0.04 = 927.6$
The investment would make \$927.60.

4. An amount \$360 are put into a savings account. The account has a 2% simple interest rate. How much money would the account have by the end of 10 years?
 $360 \cdot 10 \cdot 0.02 = 72$
At the end of 10 years, the account would have \$432.

5. How much money would an investment of \$390 in an 3-year period with a 4% simple interest rate make?
 $390 \cdot 3 \cdot 0.04 = \46.80
The investment would make \$46.80.



6. A company needed to hire a new sales manager and expected 28 people to turn in resumes for a job interview. By the end of the process, only 17 people did. What is the percentage of error?

$$\left| \frac{28 - 17}{17} \right| = \left| \frac{11}{17} \right| = 0.647 \rightarrow 64.7\%$$

The percentage of error is 64.7%.

7. What is the percent error in using $\frac{22}{7}$ as an approximation of π ?

$$\left| \frac{\frac{22}{7} - \pi}{\pi} \right| = \left| \frac{0.001264}{\pi} \right| = 0.000402499 \rightarrow 0.04\%$$

The percent error is 0.04%.

8. An estimated 1,800 people were expected to attend a rock concert at Daily's Place, but only 1,340 attended. What is the percentage of error?

$$\left| \frac{1800 - 1340}{1340} \right| = \left| \frac{460}{1340} \right| = 0.3433 \rightarrow 34.33\%$$

The percentage of error is 34.33%.

9. Justin is preparing to run a 5K race. He wanted to run 3 miles a day but, by the end of the week, he only managed to run 2 miles per day. What is the percentage of error?

$$\left| \frac{3 - 2}{2} \right| = \left| \frac{1}{2} \right| = 0.5 = 50\%$$

The percentage of error is 50%.

10. The population of a town in the year 2000 was 38,000. It was expected to rise up to 45,000 by the end of the year 2020. But, by 2020 the population was only 36,000. What was the percentage of error?

$$\left| \frac{45,000 - 36,000}{36,000} \right| = \left| \frac{9,000}{36,000} \right| = 0.25 \rightarrow 25\%$$

The percentage of error was 25%.

